Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/19/2012
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Novel Electro-hydrodynamic Technology for Gas-Liquid Phase Separation in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	05/19/2009	**End Date:**	08/19/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date now shows as 8/19/2013 per NSSC (Ed., 3/21/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	James, Conrad (Sandia National Laboratories)		
Grant/Contract No.:	NNX09AK06G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Due to the lack of gravity, the removal of bubbles formed in a flowing fluid presents a challenge in space applications. The most straightforward application of the proposed field and flow driven bubble stripping and/or two phase flow management is in lab-on-a-chip (LOC) devices envisioned for space applications. Such devices inherently have air bubbles which are difficult to remove. Existing methods of the bubble removal in LOC systems appear to be of limited use for space applications due to different functional requirements. In the first year of our work on the project, we developed an electro-hydrodynamic (EHD) method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based polymer solutions, such as saliva and urine, in a microgravity environment. The main idea behind our method is to apply an electric force directly to a droplet via capacitive coupling to external electrodes during the application of an alternating current (AC) voltage. During the second year, we developed a conveyer-belt technique that employs our EHD method for continuous		

	processing of many droplets. The third year study focused on understanding mechanisms that underlie the EHD method for separation of air bubbles from liquid. Most of experiments were conducted on suspensions of neutrally buoyant particles in order to suppress the gravity driven separation. Experiments were conducted within the range of parameters appropriate for future applications of LOC devices in microgravity. Tests on open micro-channels demonstrated that the developed EHD method can be used as an in-line debubbler in a pre-processing stage of LOC systems for bodily fluids in future space applications. A PhD student (minority) and a postdoc were involved in the research during the third year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research effort addresses the scientific and engineering issues prioritized as “Higher-Priority Areas” within the Fundamental Microgravity Research and Engineering Research Enabling Exploration programs by the Committee on Microgravity Research and as “Critical” and “Severely-Limiting” by the workshop on two-phase flow in microgravity. Due to the lack of gravity, the displacement and removal of bubbles formed in a flowing fluid presents a challenge in space applications. The inadvertent appearance and accumulation of vapor and gas bubbles in liquids can cause fluid systems, such as liquid pumps, volumetric techniques for sampling liquids, and biotechnology systems, fail to perform. The ongoing research effort should improve our understanding of the behavior of gas-liquid mixtures subject to a high-gradient electric field and facilitate the design and optimization of devices to mitigate these undesirable phenomena in microgravity. In Dec 2011 we submitted a proposal “Parabolic Flight: Validation of Electro-Hydrodynamic Gas-Liquid Phase Separation in Microgravity” to address NASA Announcement Of Flight Opportunities NOCT110. We plan to assemble the EHD system in flight-test configuration and carry out flight tests on four fluids with different properties in a weeklong campaign with 4 flights and vary an applied voltage and a bubble size from one parabola to another in a flight. Our estimates indicate that 15-20 seconds of zero gravity would be sufficient to test predictions for the bubble segregation and removal. Basic system fluidic and electric components will represent a miniaturized version of our unit for experiments on the suspension electro-hydrodynamics flown in 2003. The proposing team has sufficient expertise and resources to fabricate the system in flight-test configuration for a parabolic aircraft and carry out flight tests within the identified schedule. If successful, flight tests should validate the EHD method and provide critical data for the development of EHD devices to control, manipulate, and separate bubbles in a wide range of space technologies
Task Progress:	The study was conducted within the scope of the originally-proposed research plan. Arnon Chait, Technical Officer, discussed opportunities at GRC for applications of the proposed electro-hydrodynamic technology for gas-liquid phase separation in microgravity. The consensus was that the most straightforward application of the proposed EHD method is in lab-on-a-chip (LOC) devices envisioned for testing body fluids, such as saliva and urine, in microgravity. LOC devices inherently have air bubbles which are difficult to remove. Following this suggestion, most experiments were carried out on aqueous polymer solutions whose properties were similar to that of saliva. In the first year of our work on the project, we developed a new method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based solutions, such as saliva and urine, in a microgravity environment. The proposed method overcomes the critical challenges facing current technologies for gas-liquid separation in microgravity and appears to meet the stringent requirements of lab-on-a-chip devices for space applications. During the second year, we developed a conveyer-belt technique that employs the proposed method for continuous processing of many droplets. As air bubbles inevitably experience a buoyancy force in a normal gravity environment, we also performed experiments on two suspensions of neutrally buoyant particles to confirm that the EHD method does not require the presence of the buoyancy force exerted on air bubbles: (a) 1%-suspension of blue polyethylene microspheres dispersed in a saliva simulant and (b) 0.5% -suspensions of hollow glass particles in low volatile polyethylene glycol (PEG 200). Experiments demonstrated that the EHD method can operate over a broad range of electrical conductivity and viscosity of suspending fluids. Results of our experiments were discussed at the seminar given by PI in NASA's Glenn Research Center in Dec 2010. The third year study focused on understanding mechanisms that underlie the EHD method for separation of air bubbles from liquid. Most of experiments were conducted on suspensions of neutrally buoyant particles in order to suppress the gravity driven separation. We developed the diagram of operating modes of droplet oscillations expressed in terms of the separation between the electrodes H and the effective field strength, E, calculated as an applied voltage, U, divided by H. Using a high-speed camera, we recorded the evolution of saliva simulant and PEG 200 droplets following the application of a strong electric field. Our experiments suggest that the particle segregation is associated with the fluid flow in a droplet caused by the presence of a large electric stress normal to the liquid surface. Following discussions in NASA GRC, we conducted experiments on segregation and removal of air bubbles in open microchannels similar to LOC used for testing body fluids in microgravity. The tests demonstrated that the developed EHD method can be used as an in-line debubbler in a pre-processing stage of LOC systems for bodily fluids in future space applications. However, it remains unclear the extent to which suspensions of neutrally buoyant particles subjected to an electric field and flow can mimic the field driven behavior of bubble dispersions in microgravity. The parabolic flight tests are therefore required to provide the ultimate proof-of-concept for the EHD gas-liquid separation in microgravity and elucidate the salient effects of the buoyancy force.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Elele E, Shen Y, Khusid B. "Accumulation of Air Bubbles In Field Driven Oscillations of a Droplet" Presented at the 2011 AIChE Annual Meeting/2011 Annual Meeting of the American Electrophoresis Society (AES), Minneapolis, MN, October 16-21, 2011. 2011 AIChE/2011 Annual Meeting of the American Electrophoresis Society (AES) - Proceedings, http://www3.aiche.org//Proceedings/Abstract.aspx?PaperID=231032 , Oct-2011

Abstracts for Journals and Proceedings	Elele E, Shen Y, Khusid B. "Field-Driven Concentration and Removal of Air Bubbles from Biofluid Droplets." Presented at ASME 2011 International Mechanical Engineering Congress, The 9th Symposium on Electric, Magnetic & Thermal Phenomena in Micro and Nano-Scale Systems, Denver, CO, November 11-17, 2011. ASME 2011 International Mechanical Engineering Congress, The 9th Symposium on Electric, Magnetic & Thermal Phenomena in Micro and Nano-Scale Systems, Denver, CO, November 11-17, 2011. IMECE2011-64732, http://www.asmeconferences.org/Congress2011/TechnicalProgramOverview.cfm#613 , Nov-2011